

EcoCast: An Automated ETL Pipeline and Interactive Dashboard for Local Renewable Energy Potential

Introduction and Problem Statement

Small-scale renewable energy operators, smart-grid planners, and local sustainability decision-makers often need to understand when and where solar or wind conditions are strongest. However, weather forecast data is usually raw, fast-moving, and difficult to interpret without technical processing. Forecast data is often stored in complex API responses, which makes it challenging for non-technical users to quickly compare locations or identify useful energy-generation windows.

EcoCast addresses this problem by creating an automated data engineering pipeline that collects hourly weather forecast data, transforms it into structured datasets, stores it in a relational format, and presents the results through an interactive dashboard. The goal is to turn raw weather data into clear renewable energy insights that can help users compare local solar and wind potential across selected cities.

2. Objectives

The main objective of this project is to build an end-to-end data pipeline that supports renewable energy analytics using weather forecast data.

The project will focus on the following objectives:

Extract: Collect 7-day hourly forecast data from the Open-Meteo API using Python.

Transform: Clean, flatten, and standardize the API response into structured tabular datasets.

Load: Store the processed data in a relational database structure and create analytics-ready outputs.

Validate: Apply data quality checks for missing values, duplicate records, invalid ranges, and relationship consistency.

Analyze: Create renewable energy scoring metrics based on cloud cover and wind speed.

Visualize: Build an interactive dashboard that allows users to filter by city and evaluate solar, wind, and overall renewable readiness.

3. Data Source Plan

The primary data source for this project is the Open-Meteo API, a public weather forecast API that provides hourly forecast data without requiring an API key.

The project will use the following weather variables:

- * Temperature
- * Cloud cover
- * Wind speed
- * Wind direction
- * Forecast date and time
- * Geographic location information

These variables are relevant because cloud cover can be used as a simple indicator of solar potential, while wind speed can be used to estimate wind energy potential. The data will be collected for selected cities such as Louisville, Lexington, and Indianapolis to allow basic location comparison.

4. Methodology and Technical Approach

EcoCast will follow a standard data engineering workflow: extraction, transformation, loading, validation, and visualization.

During the extraction stage, Python will send requests to the Open-Meteo API and collect hourly forecast data for selected cities. The raw API responses will be saved for traceability.

During the transformation stage, the nested JSON data will be flattened into structured tables. Timestamps will be converted into usable date and time fields, numeric weather columns will be cleaned, and duplicate or invalid records will be removed.

During the loading stage, the project will organize the data into relational tables and produce analytics-ready CSV files for dashboard development. The database design will support multiple locations and forecast records.

During the validation stage, the pipeline will check for missing values, duplicate records, invalid weather ranges, and relationship consistency between tables.

During the visualization stage, the cleaned data will be used to create an interactive dashboard with filters, KPI cards, and charts showing weather trends and renewable energy potential.

5. Planned Database Structure

The project will use a relational database structure with three main tables:

Table Name	Purpose
locations	Stores city names, latitude, longitude, and timezone information.
weather_forecasts	Stores hourly weather forecast records for each location.
renewable_scores recommendations.	Stores calculated solar scores, wind scores, and renewable energy

This structure allows the project to separate location information from forecast data and renewable energy scoring results. It also supports scalability because more cities can be added later without redesigning the entire pipeline.

6. Project Timeline

The project will be completed in five phases that align with the course deliverables. There are as follows.

Week 1 Project Proposal and Planning

Define the business problem, identify the Open-Meteo API as the data source, outline project objectives, and design the initial pipeline concept.

Week 2 Database Design and Initial Loading

Create the database schema, document the tables and relationships, build the ER diagram, and prepare the initial database loading plan.

Week 3 will attempt to build ETL Pipeline and Data Quality

Build the Python ETL pipeline, extract weather forecast data, clean and transform records, create renewable energy scores, and run validation checks.

Week 4- Dashboard Development

Develop an interactive dashboard with visualizations, filters, KPI cards, and renewable energy insights for selected cities.

Week 5- Final Repository and Executive Presentation

Organize the full GitHub repository, finalize documentation, prepare screenshots and demo materials, and deliver a five-minute executive presentation.

This timeline is realistic because each phase builds on the previous week's deliverable. The project begins with planning, moves into data modeling and ETL development, then finishes with dashboarding, documentation, and executive communication.

7. Expected Outcomes

By the end of the project, EcoCast is expected to produce a complete end-to-end data engineering and analytics solution.

The expected outcomes include:

1. A working Python ETL pipeline that extracts hourly forecast data from the Open-Meteo API.
2. Clean and structured datasets containing temperature, cloud cover, wind speed, wind direction, and forecast time.
3. A relational database schema with tables for locations, weather forecasts, and renewable energy scores.
4. Renewable energy scoring metrics, including solar score, wind score, and an overall recommendation.
5. A validation framework that checks for missing values, duplicates, invalid ranges, row count issues, and relationship consistency.
6. Analytics-ready outputs that can be used in dashboard tools such as Dash or Power BI.

7. An interactive dashboard with city filtering, KPI cards, trend charts, and renewable energy comparisons.

8. A completed GitHub repository containing the proposal, data source plan, schema documentation, ER diagram, ETL script, SQL files, dashboard files, screenshots, README, and final presentation.

The main business outcome is to help users quickly compare renewable energy potential across selected cities and make more informed decisions using clear, visual, and updated weather-based insights.

9. Business Value

EcoCast provides value by turning raw weather forecast data into simple renewable energy insights. Instead of requiring users to manually interpret complex weather data, the dashboard presents the information in a business-friendly format.

The project can help users answer questions such as:

- * Which city has stronger solar potential based on cloud cover?
- * Which location has stronger wind potential based on wind speed?
- * How do weather conditions change throughout the forecast period?
- * Which locations may be better suited for renewable energy planning?

Although EcoCast is a student project and not an engineering-grade energy prediction system, it demonstrates how data pipelines can support practical business decision-making.

10. Tools and Technologies

The project will use the following tools and technologies:

Tool	Purpose
python	Main programming language for ETL pipeline
requests	Connects to the open meteo API
pandas	Cleans, transforms and organizes
Sql/postgresql ready structure	Supports relational data storage&querying
Dash	Provides interactive dashboard visualization
Github	Stores the final project repository and documentation

11. Conclusion

EcoCast is designed to demonstrate the full data engineering lifecycle, from data extraction to business communication. The project starts with raw weather forecast data and converts it into structured, validated, and visual renewable energy insights. By combining API extraction, ETL processing, database design, validation, and dashboarding, EcoCast provides a practical example of how data pipelines can support better decision-making for renewable energy planning.

Pipeline Diagram

The data flows sequentially through a standard 3-layer data engineering lifecycle:



